

MATCHMINDAI - PROJECT PRESENTATION DECK

Real-Time Sports Intelligence & 10,000-Run Monte Carlo Prediction Platform

SLIDE 1 OF 10 - Slide 1: Introduction & Title Overview

- **Project Name:** MatchMindAi (Real-Time Cricket Intelligence & Prediction Platform).
- **Live Web App:** <https://matchmindai-t5mgmcv6r79nflpye5ykla.streamlit.app>
- **GitHub Repository:** <https://github.com/Soham-ss/MatchMindAi>
- **Tech Stack:** Python 3.14, Streamlit, Scikit-Learn (RandomForest), Plotly, NumPy, Pandas, BeautifulSoup4, HTML5/CSS3.
- **Core Goal:** Provide cricket fans with 10,000 ball-by-ball simulated match predictions and crisp 1-line answers for all 18 IPL seasons (2008-2026).

SLIDE 2 OF 10 - Slide 2: The Problem We Solved

- **Problem 1 (Static Data):** Traditional prediction sites only look at old stats and ignore today's live pitch reports, weather updates, and stadium dew factors.
- **Problem 2 (Messy AI Responses):** General AI chatbots return long, messy URL links (Source [1]...) instead of giving simple, direct 1-line answers.
- **Problem 3 (Lack of Simulation):** Fans cannot simulate thousands of ball-by-ball match scenarios to view score distribution ranges.

SLIDE 3 OF 10 - Slide 3: Our Solution (MatchMindAi)

- **10,000-Run Monte Carlo Simulation Engine:** Simulates 10,000 ball-by-ball matches per click based on team ratings, stadium pitch profiles, and dew factors.
- **1-Line Direct Factual Answers:** Instantly answers player team queries, Orange Cap winners, Purple Cap winners, and Champions for any year (2008-2026).
- **Real-Time Internet Scraping:** Scrapes live Google News RSS feeds to incorporate today's pitch and weather news.
- **High-Energy 4D Cyber UI:** Dark mode aesthetic, live stadium video header, and interactive cursor particle depth radar.

SLIDE 4 OF 10 - Slide 4: Main Features & Capabilities

- **Smart AI Chatbot:** Recognizes user introductions personally ('Hi my name is Soham' -> 'Hello Soham! 🟩') and answers cricket queries.
- **18-Year IPL Knowledge Base (2008-2026):** Maps every Orange Cap, Purple Cap, and Champion for all 18 seasons (e.g. 'who won orange cap in 2017' -> David Warner, 641 runs).
- **Pre-2008 Boundary Check:** Catches invalid pre-2008 queries ('who won in 2005') and explains IPL was founded in 2008 with Shaun Marsh winning the first Orange Cap.
- **1-Click Match Predictor:** Pick 2 teams and a venue to get win probability meters and score histogram graphs.

SLIDE 5 OF 10 - Slide 5: System Architecture & Data Flow

- **1. User Query / Selection:** User enters a question or selects two teams.
- **2. Smart Intent Router:** Checks if query is a greeting, identity question, IPL history lookup, or match prediction.
- **3. Parallel Sub-Agents Triggered:** Web Scraper Agent + Monte Carlo Simulator + RandomForest ML Model.
- **4. Ensemble Blend:** Combines 55% Monte Carlo + 45% ML to calculate final win probability.
- **5. Streamlit Cyber UI:** Renders win gauges, score graphs, and crisp markdown commentary.

SLIDE 6 OF 10 - Slide 6: Machine Learning & Simulation Details

- **RandomForest ML Classifier:** Trained on historical IPL datasets using scikit-learn. Pre-compiled model weights are stored as .pkl files inside models/.
- **Stochastic Monte Carlo Math:** Calculates run probabilities per ball based on team ratings (CSK 88, MI 87, RCB 85) and venue dew factors (Wankhede 0.12).
- **Interactive Plotly Graphs:** Displays score distribution histograms showing expected run ranges across 10,000 simulated matches.

SLIDE 7 OF 10 - Slide 7: Technical Challenges Faced & Solutions

- **Challenge 1 (GitHub 100 MB Limit):** git push failed because dataset/IPL.csv was 104.93 MB. Solution: Removed IPL.csv from git history because model weights were already compiled into .pkl files in models/.
- **Challenge 2 (Streamlit Cloud Linux Paths):** Relative paths like 'models/match_predictor.pkl' crashed on Linux. Solution: Used os.path.abspath to dynamically locate exact folder paths.
- **Challenge 3 (Missing Packages):** missing beautifulsoup4 and python-dotenv caused ModuleNotFoundError. Solution: Added packages to requirements.txt and wrapped imports in safe try-except blocks.
- **Challenge 4 (Phrasing Variations):** 'who wins 2018 ipl' (present tense) bypassed 'who won' router. Solution: Expanded regex keywords to catch all tense variations.

SLIDE 8 OF 10 - Slide 8: UI/UX & High-Energy Design System

- **Cyberpunk Dark Aesthetic:** Premium #030712 background with HSL tailored emerald accents (#10b981, #34d399).
- **Modern Typography:** Custom Google Font (Outfit) for clean legibility across laptops and mobile screens.

- **Dynamic Visual FX:** Looping HD stadium lights video banner + interactive 4D particle cursor depth radar.
- **Sticky Bottom Chat Input:** Pinned Chat Box stays permanently visible at the bottom of the screen when scrolling long chats.

SLIDE 9 OF 10 - Slide 9: Cloud Deployment & Infrastructure

- **Live Hosting:** Deployed live on Streamlit Community Cloud with zero downtime. Accessible globally at <https://matchmindai-t5mgmcv6r79nflpye5ykla.streamlit.app>
- **Source Control:** Version-controlled on GitHub (<https://github.com/Soham-ss/MatchMindAI>).
- **Automated CI/CD:** Pushing code updates to GitHub main branch automatically triggers a rebuild on Streamlit Cloud within ~30 seconds.

SLIDE 10 OF 10 - Slide 10: Conclusion & Future Scope

- **Conclusion:** MatchMindAi successfully combines artificial intelligence, machine learning, Monte Carlo simulations, and real-time internet scraping into a fast, accurate, and stunning web platform.
- **Future Roadmap 1:** Connect live ball-by-ball score APIs for real-time win probability shifts during active matches.
- **Future Roadmap 2:** Fantasy cricket team optimizer for Dream11 / My11Circle lineups.
- **Future Roadmap 3:** Multi-sport expansion to Football (Premier League, Champions League) and Basketball (NBA).